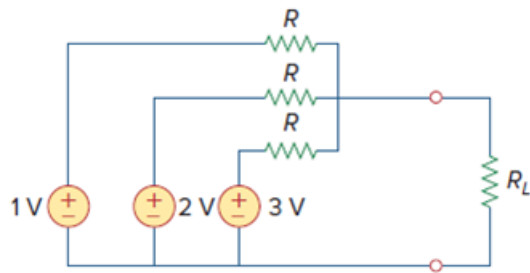


Problem

For the circuit in Fig. 4.141, determine the value of R such that the maximum power delivered to the load is 12 mW.

Figure. 4.141:



Step-by-step solution

Step 1 of 5

Refer to Figure 4.141 in the text book.

Disconnect the load resistance to calculate the Thevenin's voltage.

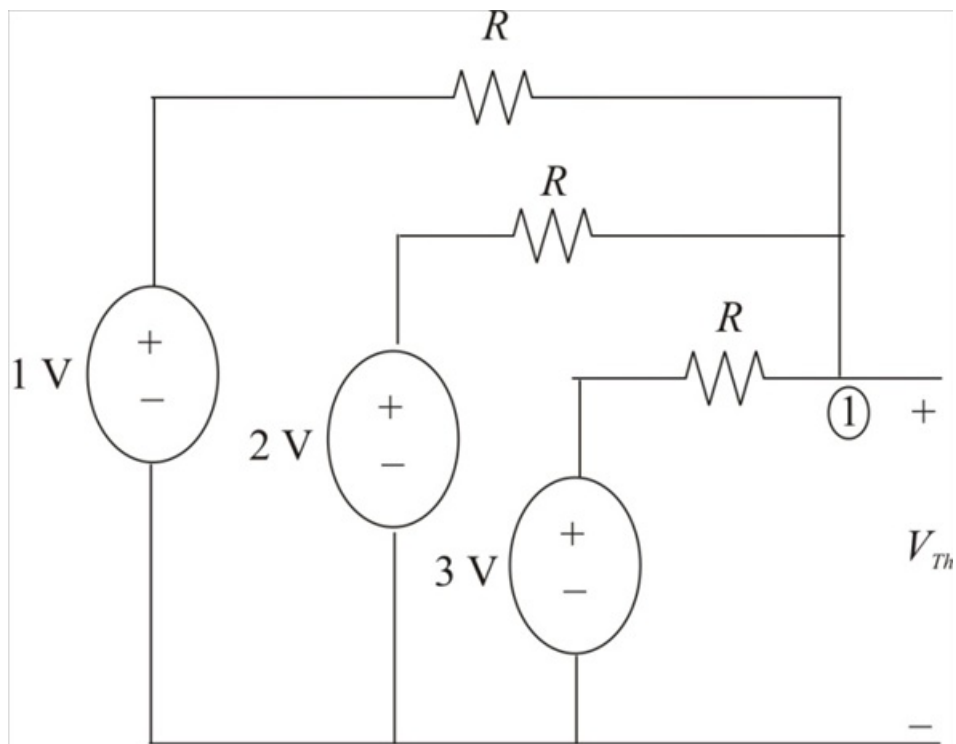


Figure 1

Step 2 of 5

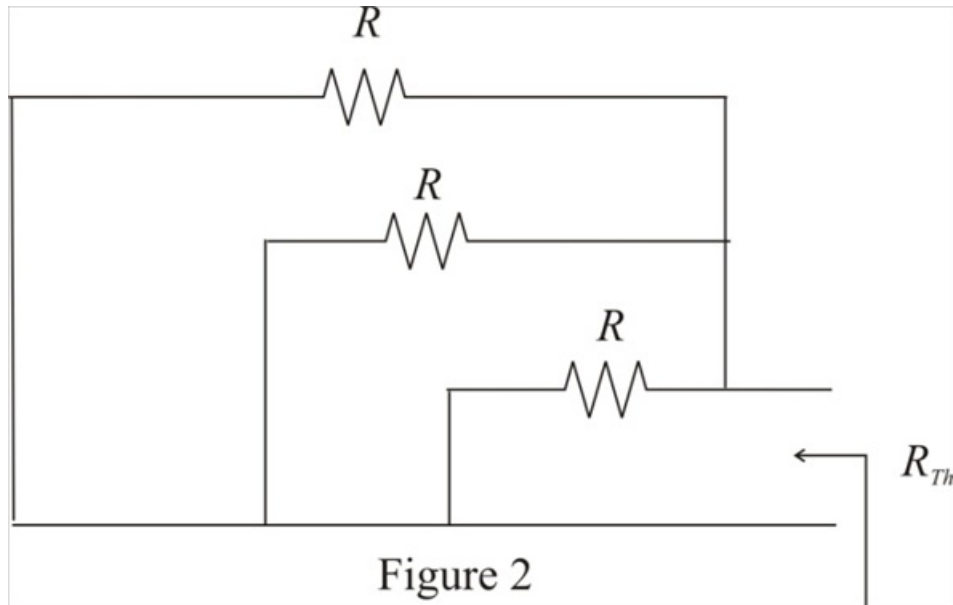
Apply Kirchhoff's current law to node-1.

$$\begin{aligned}\frac{V_{Th}-1}{R} + \frac{V_{Th}-2}{R} + \frac{V_{Th}-3}{R} &= 0 \\ V_{Th} \left(\frac{1}{R} + \frac{1}{R} + \frac{1}{R} \right) - \left(\frac{1}{R} + \frac{2}{R} + \frac{3}{R} \right) &= 0 \\ V_{Th} \left(\frac{3}{R} \right) - \left(\frac{6}{R} \right) &= 0 \\ V_{Th} \left(\frac{3}{R} \right) &= \left(\frac{6}{R} \right) \\ V_{Th} &= 2 \text{ V}\end{aligned}$$

Therefore, the value of the Thevenin's voltage V_{Th} is, 2 V.

Step 3 of 5

Set the values of the sources to calculate the value of the Thevenin's resistance.



Step 4 of 5

Calculate the value of the Thevenin's resistance R_{Th} .

$$\begin{aligned}R_{Th} &= R \parallel R \parallel R \\ &= \frac{(R)(R)(R)}{(R)(R) + (R)(R) + (R)(R)} \\ &= \frac{(R)(R)(R)}{3(R)(R)} \\ &= \frac{R}{3}\end{aligned}$$

Therefore, the value of the Thevenin's resistance R_{Th} is, $\frac{R}{3}$.

For the maximum power to be delivered to the load, the value of the load must be equal to the Thevenin's resistance R_{Th} . Therefore, the value of the load R_L is, $\frac{R}{3}$.

Step 5 of 5

Calculate the maximum power absorbed by the load.

$$P_{\max} = \frac{V_{\text{Th}}^2}{4R_L}$$

Substitute 2 V for V_{Th} , 12 mW for $P_{\max}$ and $\frac{R}{3}$ for R_L .

$$12 \text{ m} = \frac{(2)^2}{4\left(\frac{R}{3}\right)}$$

$$12 \text{ m} = \frac{(3)(4)}{4R}$$

$$12 \text{ m} = \frac{12}{4R}$$

$$R = \frac{1}{4 \text{ m}}$$

$$R = 250 \, \Omega$$

Therefore, the required value of load resistance is $250 \, \Omega$.